

# AI SIGN LANGUAGE TO SPEECH CONVERTER

A Project Report

*Submitted by*

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in partial fulfillment for the award of the degree of

**BACHELOR OF TECHNOLOGY**  
**IN**  
**COMPUTER SCIENCE & ENGINEERING**  
(Specialization in Artificial Intelligence and Machine Learning)



Centre for Artificial Intelligence and Machine Learning  
Department of Computer Science and Engineering  
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Institute of Technical Education and Research

**Siksha 'O' Anusandhan**  
**(Deemed to be University)**

Bhubaneswar, Odisha, India

June, 2026

# CERTIFICATE



This is to certify that the project report titled “**AI SIGN LANGUAGE TO SPEECH CONVERTER**” being submitted by **Saurav Kumar Jena , Ashutosh Sahoo , Bishwa Ranjan Mishra , Siddhant Mahapatra** of **Section - 12** to the **Institute of Technical Education and Research, Siksha ‘O’ Anusandhan (Deemed to be University)** , Bhubaneswar for the partial fulfillment for the degree of **Bachelor of Technology in Computer Science and Engineering (Specialization in Artificial Intelligence and Machine Learning)** is a record of original confide work carried out by them under our supervision and guidance. The project work, in my opinion, has reached the requisite standard fulfilling the requirements for the Degree of **Bachelor of Technology**.

The results contained in this project work have not been submitted in part or full to any other University or Institute for the award of any degree or diploma.

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# ACKNOWLEDGEMENT

We really want to thank everyone who helped and supported us as we finished our Senior Year Project, “AI Sign Language to Speech Converter.”

First off, a huge thanks to our project guide. She gave us great advice, kept us motivated, and offered helpful feedback every step of the way. Her encouragement and guidance truly pushed us to do much better with both the technical parts and the documentation for this project.

We also want to thank our institution for giving us the chance to work on such an interesting, though technically challenging, project that involved using AI.

Then, to our team members—thank you for all your hard work and for cooperating so well throughout the entire project.

Finally, we’re grateful to our parents, friends, and everyone else who cheered us on and kept us going while we were working on this project.

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# DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/fact/source in our submission. We understand that any violation of the above will cause for disciplinary action by the University and can also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been taken when needed.

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# REPORT APPROVAL

This project report titled “**Ai Sign Language To Speech Converter**” submitted by **Saurav Kumar Jena , Ashutosh Sahoo , Bishwa Ranjan Mishra , Siddhant Mahapatra** is approved for the degree of **Bachelor of Technology in Computer Science and Engineering (Specialization in Artificial Intelligence and Machine Learning)**.

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Chief Coordinator, CAIML

# PREFACE

We chose a Project was “AI Sign Language to Speech Converter”. Its primary goal was to diminish the communication barrier between ordinary people and hearing , speech disabled individuals. We know sign language is a very crucial communication skill , many times the lack of knowledge of sign language creates a barrier, and this is where our algorithm plays a role. Our objective was to build an algorithm that would recognize the sign movements and translate them into spoken language.

We knew that several algorithms already existed, but our primary focus was on making the system lightweight and efficient using a Random Forest Classifier along with spelling correction so that accurate text and corresponding speech output could be obtained. As a result , this project is reliable because of its efficiency and performance.

From this project, we learned many concepts related to Artificial Intelligence , Machine Learning , Computer Vision , gesture recognition , datasets, training , and prediction processes. We also gained practical knowledge about implementing machine learning algorithms.

Finally, we would like to thank our project guide and supervisor for their valuable guidance , encouragement , and support throughout the completion of this project.

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# Chapter 1

## INTRODUCTION

### 1.1 Introduction

As technology advances, artificial intelligence and machine learning are transforming communication accessibility. One of the most significant applications of this technology is in sign language recognition, which will help close the communication gap between those who have hearing impairments and those who do not use sign language.

For many people who are deaf or have speech impairments, their primary means of communication is through sign language. Unfortunately, due to the fact that very few people actually use sign language, this can create major barriers to everyday communication. In response to these barriers, researchers are now creating intelligent systems that can recognize the user's hand gestures and convert them to some other form of output that would be understandable by someone who does not know how to sign.

Recent advances in computer vision, as well as machine learning and technology have created an opportunity to develop real-time gesture recognition systems using standard webcams and without requiring the purchase of expensive hardware. This will allow for a more cost effective, efficient and user-friendly means of providing speech from any electronic device instead of traditional means that use sensors.

Our developed project, "AI Sign Language to Speech Converter," will capture and recognize ISL hand gestures using the MediaPipe hand landmark extraction approach and use a random forest classifier to recognize the hand gestures. The output of the gesture recognition process will be converted into text and then enhanced by automatic correction of spelling errors and next word predictions. Lastly, the resulting text information will be translated into odia and will be processed to generate a speech output using a text-to-speech system. Overall, this method will produce a system that is lightweight, portable, and would be suitable for use by anyone who communicates using ISL.

## 1.2 Project Overview / Problem Statement

Living a full life requires communication, and many people can see that having difficulty with hearing or speech prevents them from communicating normally with people who do not know sign language. For example, if you were to go to school, a doctor's office, work, or to use a public service that requires communication, it would cause significant issues without any proper way to communicate, which can lead to a feeling of isolation or being misunderstood for many people.

Although sign language is a great way to communicate with people who know it, not many people outside of that group know about it. There are currently systems in place that attempt to translate sign language into a spoken language; however, those systems are typically expensive, require specialized equipment, and are typically not easy to operate, making usage on a daily basis impossible. Many of those systems limit, or do not work well in real-time, making them even more difficult to use.

Our AI Sign Language to Speech Converter project will create a new affordable and user-friendly solution that converts sign language into spoken language using AI, computer vision, and machine learning, and works in real-time. The system works by utilizing a standard camera to detect hand gestures, then uses an AI classification system known as Random Forest Classifier to identify the hand gestures. After a person has been identified through an appropriate method, gestures will be able to be recognized as understandable text and written or oral communications.

Part of the uniqueness of the project is that it incorporates a feature for correcting spelling errors. This will help to provide improved accuracy and reliability in the prediction of outputs. In contrast to older methodologies, this system was designed for ease of use, speed of operation and low-cost with no need for additional equipment to perform the same tasks; therefore, making them easy to obtain.

The ultimate goal of this project is to give those people who are deaf and/or have a speech impairment improved means of communicating, increasing access to other activities, and providing a greater sense of inclusion in our society.

## 1.3 Motivations and Objectives

We began this project to improve communication between those with a hearing impairment or who have trouble speaking, and persons that do not know sign language. Communication is fundamental today, whether in a school, hospital, workplace or just to socialize; many use social networks or other media. However, individuals with hearing impairment, or a serious hearing impairment, often find it difficult to effectively communicate since so few people globally use sign language.

Another factor as to why we chose to begin this project is that existing systems of translating sign language don't work well. The majority of current systems require expensive equipment such as gloves and/or complicated setups that are not easily accessible or usable by the majority of the population. Therefore, we feel there is a need for a simple, inexpensive and fast solution that utilizes everyday cameras and devices already owned by the majority of the population.

The end goal of this project is to create an AI system that accurately recognizes hand gestures associated with sign language and instantly transcribes them into spoken words, written words or both. To do this we will utilize a Random Forest Classifier to categorize hand gestures. Additionally, we have built-in support for spell check to maintain the best quality of output words possible.

## 1.4 Scope and Significance of the Work

The aim of this product is to create an intelligent solution that can interpret sign language from gestures in real time using machine vision and machine learning (ML). To accomplish this, an intelligent Random Forest Classifier will be utilized in conjunction with basic vision and ML algorithms to effectively track and process the gesture data into a readable character set (text) and audible speech. The design goal is to create a low-cost, lightweight, and effective system that will use typical personal computers rather than expensive hardware (as many previous systems have done) in order to perform this task.

The ultimate goal in developing this product is to help bridge the gap in communication that exists between individuals that are Deaf or Hard of Hearing (D/HH) and those

who do not know Sign Language. Communication difficulties are often what impede D/HH individuals from achieving their goals in everyday situations such as: school, work, medical services, and socializing. By converting gestures from Sign Language into clearly articulated speech, this intelligent system will assist D/HH individuals by providing a means for them to communicate more independently while also increasing accessibility and inclusivity.

A further main component of this initiative is the inclusion of an additional level of spell checking which will produce more accurate output for translations. Many future enhancements could take place such as other language support, installation on mobile devices, detection of whole sentences as opposed to single gestures, and Cloud Computing usage. Above all these examples demonstrate the ways in which artificial intelligence can be applied towards real-world problems regarding communication.

## 1.5 Organisation of the Report

Here's how the report is structured so you can understand what is going to happen as we go. As each chapter is a subdivision of the initial project, you can expect to find references/appendices at the end of each chapter.

**Chapter 1** Will contain an overview of the project (what the proposal is), background materials related to sign language, a description of why we need AI to help us communicate, a description of what the project addresses, an explanation as to why this is the chosen project, an overview of project goals and scope vs. significance of the application developed in this project.

**Chapter 2** Will layout materials that were reviewed during the literature survey concerning artificial intelligence, machine learning, computer vision, and existing systems designed to recognize sign languages. This will include reviews of gesture recognition systems that had previously been developed; an overview of how they translate signs in real time; and finally, what did not work when looking at existing methods which ultimately led us to develop a lightweight version.

**Chapter 3** Will explain how we did the job through a description of the different methods and materials that were used during the course of development with emphasis

on our work with developing a data set (images), pre-processing that was performed on those images (e.g., to assist in the overall gesture recognition workflow), development of the random forest classifier, and some ways we made our system correct for any spelling errors. The chapter will be completed with discussion about the architecture of our system, software tools, technologies used to develop the application, etc.

**chapter 4** Deals with the implementation of the project, experimentation with it and documenting the findings. The chapter outlines the specifications of the system, the details of how it was implemented, images of the completed system, results from training and prediction using the model, performance analysis and evaluation of accuracy for the entire system.

**chapter 5** Describes the practical implications of the developed system, its usability and limitations. In particular, the chapter discusses how important it is for hearing and speech impaired individuals to have access to communication; the benefits of using lightweight AI systems; and what challenges were experienced throughout implementation.

**chapter 6** Summarises everything that has been accomplished through the project, notes possible future enhancements (including multilingual to speech conversion, sentence level recognition, mobile app integration and enhanced real-time performance) and draws an overall conclusion from the project as it stands at present day.

Following the main chapters, the References section includes all of the research papers, articles, datasets, or technical resources used in developing the project. The Appendix section includes all supporting documentation that includes the project proposal, a plagiarism report, information about contributions made by each team member, images of the completed system and the source code used to build it.

# Chapter 2

## REVIEW OF LITERATURE

### 2.1 Review of Existing Systems / Related Work

There are number of studies and systems for sign language recognition that have utilized Artificial Intelligence, Machine Learning and Computer Vision technologies. Early systems mostly focused on capturing hand gestures using sensors or hardware devices so results were very expensive, not portable, and difficult to use in "real time".

Later on, researchers utilized cameras to capture sign language by using image processing and machine learning algorithms to detect the sign language gestures. According to Camgoz et al. (2018), an End-to-End Natural Sign Language to Spoken Language Translation Framework was proposed that translated sign language videos to produce spoken language sentences using the RWTH-PHOENIX-Weather-2014T dataset; But the system has low BLEU precision, grammatical errors, and is data dependent.

Other systems Which helped to improve the recognition performance were Transformer based like Sign Language Transformers Multi-Channel Transformers , that are utilize the hand, face, and body to provide better semantic recognition and therefore do not require manual alignment;

Systems were developed in recent years like SignBERT, Cross-Modal Knowledge Distillation , that have used self-supervised learning and linguistic knowledge for improved accuracy at recognizing the gestures. Others are using the practical applications from sign language(s) to create a text and/or speech output mode based on gesture(s) in real time for communicating with one another using various types of processing technologies. Many current systems are also very computationally intensive, require proprietary hardware to work properly, and have proven to be very difficult to implement efficiently in lightweight environments.

Table 2.1: Literature Survey on Sign Language Recognition Systems

| Technique                          | Year | Purpose                                        | Advantages                                   | Limitations                          |
|------------------------------------|------|------------------------------------------------|----------------------------------------------|--------------------------------------|
| Natural Sign Language Translation  | 2018 | Translate sign video into spoken language      | First end-to-end SLT framework               | Low BLEU accuracy and grammar errors |
| Sign Language Transformers         | 2020 | Joint sign recognition and translation         | Improved performance and no manual alignment | Heavy computational cost             |
| Multi-Channel Transformer          | 2020 | Capture hand, face, and body motion            | Better semantic understanding                | Complex preprocessing                |
| SignBERT                           | 2021 | Self-supervised sign representation learning   | Works well on small datasets                 | Requires pretraining stage           |
| Cross-Model Knowledge Distillation | 2024 | Improve recognition using linguistic knowledge | Improves contextual accuracy                 | High training complexity             |
| Sign Language Recognition System   | 2019 | Convert hand gestures into text                | Practical real-time implementation           | Limited dataset size                 |
| Speech-to-Text-to-Sign Translator  | 2022 | Convert speech into sign language              | Enables two-way communication                | Complex pipeline                     |
| ASL to Speech Translation          | 2024 | Convert sign language into speech              | High accuracy reported                       | Hardware dependent                   |

## 2.2 Research Gaps and Problem Identification

Most existing sign language systems utilize either deep learning or transformer-based system architectures. These types of systems necessitate a significant amount of data, substantial computations, specialized, high cost hardware/technology, and sophisticated pre-processing techniques; therefore, they cannot be utilized in lightweight/real-time environments.

Furthermore, many of the systems are predicated on the use of sensor gloves or wearable devices for gesture detection, which increases both costs and accessibility for the average user. Some vision-based systems are incapable of real-time processing and are therefore negatively affected by their susceptibility to noise in the environment, as well as background noise placement, and/or produce inaccurate prediction results for continuous gestures.

Another common gap in the existing systems is the lack of efficient spelling correction and text refinement methods. As a result of these types of errors, that affect the communication often results from incorrect gestures . Many of the advanced systems have excessive computational requirements making them inappropriate for low-resource devices.

The purpose of this project is to eliminate the existing gap in research by developing an efficient and low weight Sign Language Recognition System using Random Forest Classifier, intended for use with standard video cameras - no special purpose hardware required. Spelling Correction techniques will be integrated to enhance the accuracy of text produced by the system and improve the reliability of speech produced by the system. The principal goal of the project is to create a cost-effective, real-time communication means for individuals who are hearing impaired or who have speech impairments difficulties communicating.

# Chapter 3

## MATERIALS AND METHODS

### 3.1 Dataset Description / / Input Specification

To implement the AI sign language to speech converter, we need an organized dataset to train and test the gesture detection model. We created our own custom dataset which contain the images of hand gestures representing all alphabets and numerical symbols used in sign language. And that are indian sign.

The dataset includes 36 folders representing each of the gesture classes based on the digits 0–9 and letters A–Z. Each folder contains about 20 images each of the respective hand gesture captured in various positions and lighting, using a regular monitor camera. This was done to ensure that the system would perform well in real-world settings without requiring any special hardware devices for operation.

The purpose of the collected dataset is to train the random forest algorithm for showing accurate recognition of the hand gestures. During this process, we resized each image and extracted landmark-based features to use for classifying the gestures. In addition, we have organized the dataset in a way that allows for easy identification and processing of each gesture class during model training and prediction.

Creating the custom dataset has enabled us to have more control over how and when we collect data, how we label gestures, and how well we can perform the system. This will also result in improved accuracy during real-time recognition of gestures for lightweight systems to translate sign language.

The main input of the new system will be images that represent hand gestures from a camera. All images that have been collected will be grouped into multiple categories based on their numerical digit (0-9) or alphabetical character (A-Z).

| Parameter              | Description                                    |
|------------------------|------------------------------------------------|
| Dataset Name           | Custom Sign Language Gesture Dataset           |
| Dataset Type           | Image-based Hand Gesture Dataset               |
| Number of Classes      | 36 Gesture Classes                             |
| Gesture Categories     | Digits (0-9) and Alphabets (A-Z)               |
| Images per Class       | Approximately 20 Images                        |
| Total Images           | Approximately 720 Images                       |
| Data Collection Method | Captured using webcam                          |
| Data Nature            | Image-based Gesture Data                       |
| Model Used             | Random Forest Classifier                       |
| Purpose                | Hand Gesture Recognition and Speech Conversion |

Table 3.1: Custom Sign Language Gesture Dataset Description

The types of input used by this system can be categorised into three general categories:

- Hand Gesture Image Inputs: Includes images of the hand gestures for representing the digit (0-9) and letter (A-Z) used to recognize and classify gestures.
- Visual Feature Input: Includes important landmark and relative positions used to extract key features from the hand gesture image before performing a prediction based on the visual feature technique.
- Text Correction Input: Includes correcting spelling errors from predictive gestures with the use of spelling correction techniques (to improve the accuracy and reliability of text or speech output).

In addition to the above, these categories can all be classified into one large category 'gesture'. Using images of hand gestures and extracting visual feature inputs provides the capability for doing real time recognition of sign language compared to traditional methods which require use of expensive hardware equipment.

## 3.2 System Architecture / Block Diagram / UML Diagram / DFD / Use Case Diagram

The developed AI Sign Language to Speech Converter system connects multiple interconnected modules to achieve real-time gesture recognition, text refinement, Odia translation, and speech generation.

The process starts with the user making a sign in front of a webcam, which captures the video image. Using the MediaPipe Hands framework, the system captures the hands to extract 21 distinct reference points or landmarks. Then the extracted landmark coordinate positions are normalized into feature vectors which are delivered to a Random Forest classifier to perform gesture recognition.

Once the gestures are recognized then it is converted into text string. These outputs are also processed through SymSpell spell checker alongside a next-word prediction algorithm. The completed text is then translated to the Odia language, which the user will view in the Odia language's script, and further processed into audible speech using the Google Text-to-Speech gTTS engine.

Overall, the architecture of the system is designed to be lightweight, efficient in terms of computation time and required resources, and can be executed in real-time on standard CPU without requiring expensive equipment such as GPUs.

### **3.3 Methodology / Algorithm / Model Description**

The developed AI Sign Language to Speech Converter includes a real-time gesture recognition and communication pipeline. A standard webcam continuously captures video frames of hand gestures, feeding them directly into the MediaPipe Hands framework. This framework identifies and extracts 21 specific hand landmarks corresponding to the palm and finger joints.

Before classification, the system normalizes these extracted landmarks, transforming the raw coordinate points into one-dimensional feature vectors. These output feature vectors are input into a Random Forest classifier to predict gestures. The predicted gesture labels are then converted into text and further processed for spell-checking by using a module SymSpell and also include a prediction component for the next word.

Once the output is completely translated into Odia language, then the final output will have been converted into an audio output by the Google Text-To-Speech (gTTS) engine. This method allows an individual with hearing impairment to communicate effectively.

### 3.3.1 Algorithm

- Capture real-time video frames using a webcam.
- Detect hand regions using MediaPipe Hands.
- Extract 21 hand landmark coordinates from the detected hand.
- Normalize the landmark coordinates and generate feature vectors.
- Provide the feature vectors to the Random Forest classifier.
- Predict the corresponding sign language gesture.
- Convert the predicted gesture into text output.
- Apply spell correction and next-word prediction.
- Translate the corrected text into the Odia language.
- Convert the translated text into speech using the gTTS engine.
- Display the final text and generate speech output in real time.

### 3.3.2 Model Description

Normalized hand landmark coordinates are extracted from the MediaPipe Hands framework. Then create the Random Forest Classifier Model for recognizing hand gesture. The Random Forest classifier uses 21 hand landmarks to represent the gesture and stores the x/y coordinates of the 21 landmarks in a numeric feature set.

A Random Forest Model consists of multiple decision trees that combine to classify hand gestures via an ensemble learning approach. To verify the gesture classification during prediction, each of the individual decision trees predicts gesture class. The final classification result is based on the majority vote from all the trees within the Random Forest.

We choose the Random Forest classifiers for this project because of its comparatively high prediction accuracy, light-weight computations, fast training time, and suitability for real-time execution using normal CPU processors.

Real-time acknowledgment of gestures via MediaPipe-based landmark extraction combined with Random Forest classification yields the accurate performance level required for executing real-time gesture recognition without requiring intensive computational resources or specialized hardware components.

### **3.4 Tools, Technologies and Frameworks Used**

The AI Sign Language-to-Speech Converter uses computer vision, and machine learning programs to identify gestures in real-time and convert them to speech. Python 3.10 was used as the main programming language and the complete system was built on this programming language. OpenCV was used for capturing images from the camera and analyze those images. MediaPipe Hands was used to detect and extract 21 points of the hand that represent different locations on your hand in order to identify gestures in real-time.

The Random Forest classifier was implemented using Scikit-learn to classify the gestures. NumPy was utilized for numerical calculations and manipulating the feature vector that was extracted from the hand. SymSpellPy was used to perform automatic correction of misspelled words and to predict the next word in a given sentence. Deep translator was used to convert the recognized text from the sign language to odia. Google Text to Speech (gTTS) was used for generating the speech output. Playsound was used for playing the audio output of the generated speech. PyCharm IDE was used for creating and testing the program.

### **3.5 Performance Metrics / Evaluation Criteria**

The system we developed has been tested for its performance based on several metrics that assess accuracy of gesture recognition, stability of prediction, and efficiency of execution in real-time. Accuracy is the percentage of gestures classified correctly by the Random Forest Classifier; precision, recall, and F1-Score are used to evaluate the reliability of the classification and the overall performance of the model itself.

The accuracy of classifier predictions was assessed through the use of confusion matrices. Confusion matrices allow for visualisation of the gestures that were predicted correctly and incorrectly, and also provide a way to identify confusion within visually similar signs.

To thoroughly assess the proposed system, we measured its real-time operational speed alongside its overall accuracy and robustness. This involved benchmarking the exact time required to detect a gesture, output the initial text, translate that text, and finally synthesize the spoken audio. We also gauged the effectiveness of the gesture stabilization mechanism, specifically looking at how well it prevented accidental duplicate characters when a user moved their hands continuously. Ultimately, the system delivered high-fidelity, real-time processing on conventional CPUs, locking in a gesture classification accuracy of over 99%.

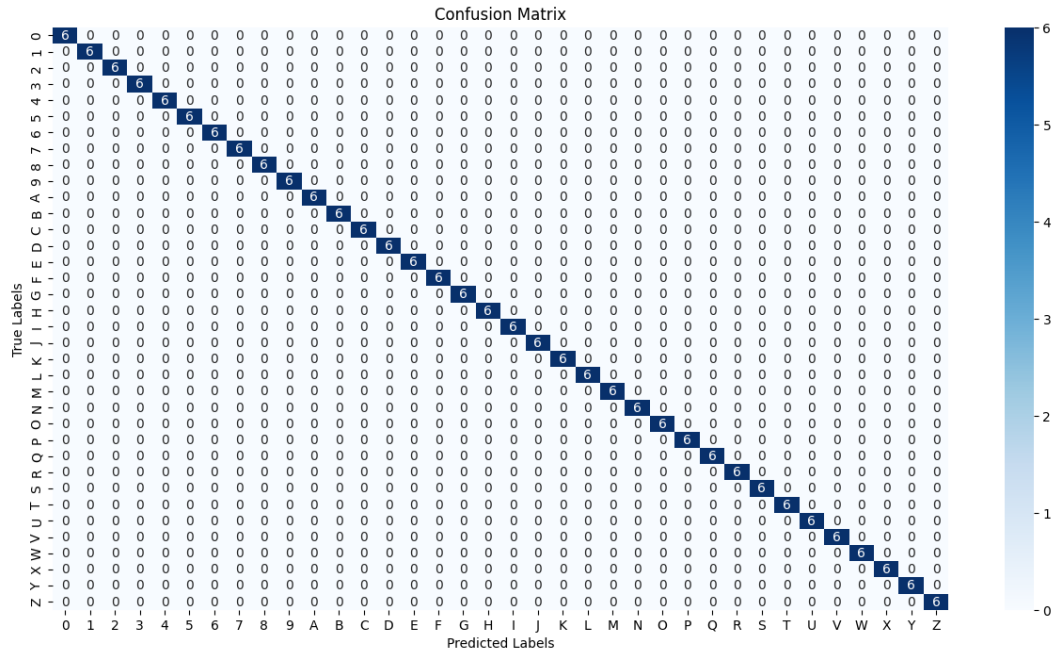


Figure 3.1: Performance Matrix

# Chapter 4

## EXPERIMENTATION AND RESULTS / APPLICATION AND SYSTEM DESIGN

### 4.1 System Specifications / Experimental Setup

This AI-powered Sign Language to Speech Converter system was implemented and evaluated on a standard CPU-based environment, without using specialized GPU. The software was built using Artificial Intelligence, Machine Learning, Computer Vision, and Natural Language Processing packages. Then the MediaPipe was used for the continuous capture of real-time gesture data with a webcam and extracting hand landmarks from the captured data. The extracted coordinate position of the landmarks were converted into normalized feature vectors that use in the classification of gestures with a Random Forest classifier to provide accurate gesture recognition.

The experimental status is described in Table 4.1.

| Specification             | Details                                         |
|---------------------------|-------------------------------------------------|
| Processor                 | Intel Core i5 / AMD Ryzen Equivalent            |
| RAM                       | 8 GB                                            |
| Operating System          | Windows 10/11                                   |
| Programming Language      | Python 3.10                                     |
| Development Environment   | PyCharm IDE                                     |
| Libraries Utilized        | OpenCV, MediaPipe, Scikit-learn, SymSpell, gTTS |
| Input Device              | Webcam Camera                                   |
| Classification Model      | Random Forest Classifier                        |
| Feature Extraction Method | MediaPipe Hand Landmarks                        |
| Speech Engine             | Google Text-to-Speech (gTTS)                    |
| Translation Engine        | Deep Translator                                 |
| Gesture Classes           | Alphabets (A–Z) and Digits (0–9)                |

Table 4.1: Experimental Setup and System Specifications

## 4.2 Parameter Settings / Configuration Details

The developed system utilizes several configurable parameters to find accurate real-time gesture recognition, efficient text processing, and reliable speech generation.

Table 4.2 outlines the key configurations

| Parameter                      | Value           |
|--------------------------------|-----------------|
| Number of Hand Landmarks       | 21              |
| Total Features per Gesture     | 42              |
| Classifier Algorithm           | Random Forest   |
| Number of Trees                | 100             |
| Maximum Hands Detected         | 2               |
| Detection Confidence Threshold | 0.5             |
| Tracking Confidence Threshold  | 0.5             |
| Gesture Hold Duration          | 10 Seconds      |
| Prediction Buffer Size         | 15 Predictions  |
| Spell Correction Method        | SymSpell        |
| Speech Generation              | gTTS            |
| Translation Language           | English to Odia |

Table 4.2: Parameter Settings Used in the Proposed System

## 4.3 Implementation Details / Module Description

The system is the combination of multiple interconnected modules that simultaneously enable intelligent and real-time sign language communication.

### 4.3.1 Hand Landmark Detection Module

Webcam captures live video frames and processes them with the MediaPipe Hands Framework by providing 21 landmarks for the joints of the finger or location of the fingers along with the palms. Normally these landmarks are normalized using their minimum X and Y values to ensure consistency across hand motion and size scaling as they are processed. Using this method it reduces the computational energy associated with using raw images when using deep learning algorithms.

### 4.3.2 Gesture Classification Module

The normalized landmark coordinates are transformed into 42 dimensional feature vectors and classified using a Random Forest model. This classifier predicts gesture classes (A–Z,

0–9) with high accuracy and minimal latency and often with specific outputs.

### **4.3.3 Stable Gesture Typing Module**

To mitigate noisy predictions and repeated character generation, a gesture stabilization mechanism was implemented, which detects the hands and holds for a 6 second time interval. Characters are displayed only if the predictions continue to be consistent for a specified amount of time. Prediction history filtering further enhances stability by selecting the most reliable predictions from recent buffers.

### **4.3.4 Spell Correction Module**

The generated text undergoes dictionary based spell correction using SymSpell method . This gives us improved sentence quality and communication reliability by correcting the uncorrected characters.

### **4.3.5 Next Word Prediction Module**

A lightweight next word prediction module suggests more context aware word completions based on predefined and wanted phrase patterns and frequently used vocabulary, thereby improving typing speed and next predicting hussels.

### **4.3.6 Multilingual Transliteration Module**

The predicted English text is transliterated into Odia for regional accessibility. For example:

HELLO → Namaskara

This transliteration avoids Unicode rendering issues and it doesnt support the authentic odia language support in OpenCV while enhancing usability.

### **4.3.7 Speech Synthesis Module**

Text outputs are translated into Odia and subsequently synthesized into speech using the Google Text-to-Speech (gTTS) engine, which generates natural Odia audio output. Not only does this ensure accuracy in linguistic translation but also facilitates seamless communication in real time in bilingual settings. By bridging gesture-based input with

spoken Odia, the system enables inclusive interaction for regional language users, thereby enhancing accessibility and practical usability in everyday communication scenarios.

### **4.3.8 Interactive Smart Keyboard Interface**

The system integrates a smart keyboard interface with interactive controls such as:

- Speak Button
- Clear Button
- Stop Button
- Clickable Suggestions
- Gesture Hold Status Display with a timer count
- Prediction Display

This interface ensures seamless user interaction with the communication system.

## **4.4 Results and Outputs**

The system successfully demonstrated real-time gesture recognition, text generation, transliteration, and speech synthesis. Key observations include:

- High accuracy while recognize the gesture
- Stable typing performance with reduced noise
- Effective spelling correction of word
- Context-aware next-word prediction
- Real-time Odia language conversion
- Bilingual speech synthesis
- Low-latency CPU-based execution

The Random Forest classifier achieved approximately 99% accuracy during evaluation. Gesture stabilization mechanisms significantly reduced accidental character generation.



Figure 4.1: Gesture Recognition and text Generation Output

## 4.5 Result Analysis and Validation

The AI-powered multilingual gesture keyboard demonstrated efficient real-time performance on standard CPU hardware without GPU acceleration. MediaPipe landmark extraction produced lightweight, precise features for classification. The Random Forest classifier ensured fast inference with reduced complexity compared to deep learning alternatives.

Prediction history filtering and gesture stabilization improve the system's robustness against transient hand movements. SymSpell-based spell correction enhanced communication quality, while next-word prediction reduced typing effort. Multilingual transliterations and bilingual speech synthesis extend the accessibility of the system and have created a system that can be utilized in practical applications.

Overall, the proposed system achieved lightweight, real-time Indian Sign Language recognition with intelligent communication assistance, validating its suitability for practical deployment.

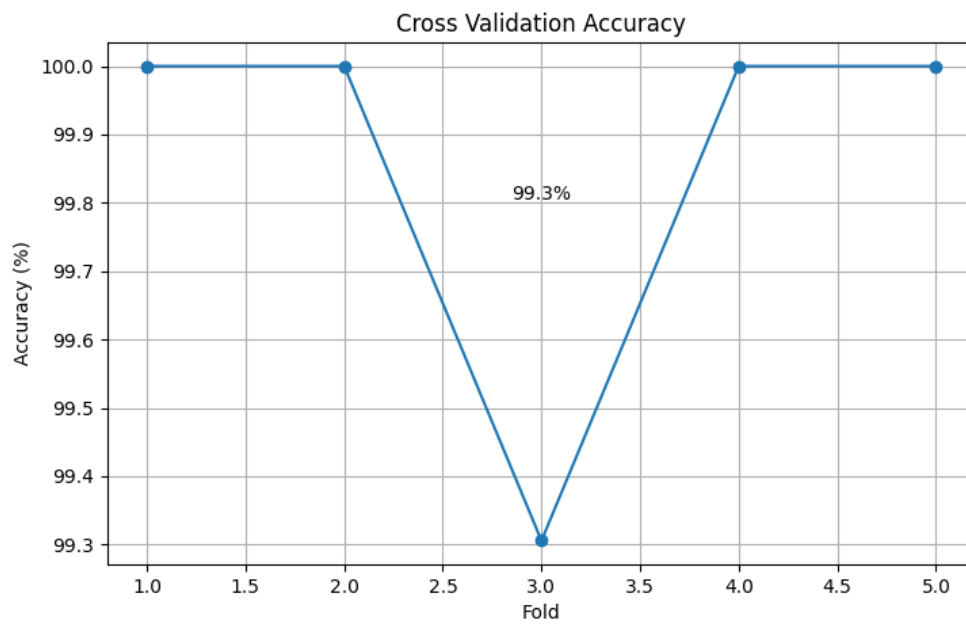


Figure 4.2: Cross Validation Graph

# Chapter 5

## ETHICAL, SOCIAL & SUSTAINABILITY CONSIDERATIONS

### 5.1 Ethical Consideration

This system has been created for the purpose of enabling hearing-impaired people to communicate effectively with those who don't have hearing impairments. The AI Sign Language to Speech Converter does not collect or save any confidential personal data. It protects user privacy while keeping their data and securely stored.

Gesture recognition for this tool operates in real-time via a webcam; there is no requirement to store biometric data through external wearable technology or devices.

We designed this AI Sign Language to Speech Converter to foster inclusive communication, ensuring fair access for individuals regardless of language barriers or hearing impairments. Every phase of development prioritized generating accurate, context-appropriate text and vocal outputs for everyday interactions. However, because the system relies on machine learning predictions, absolute accuracy cannot be guaranteed, and occasional gesture misclassifications may happen. Therefore, we advise against using this software in critical situations where absolute communication precision is mandatory.

### 5.2 Social Impact and Societal Relevance

The proposed system has the potential to improve the social interaction between hearing-impaired people and others who do not use sign language, by converting sign language gestures into text and audible speech so that people can understand each other. It provides facilities in communication within schools, work settings, health care and public settings.

Integrating speech synthesis alongside bidirectional translation for regional languages like Odia substantially enhances accessibility for local communities. This inclusion ensures the system remains culturally relevant while meeting the specific communicative needs of the region. Furthermore, the lightweight, cost-effective architecture allows the software to execute on standard commercial-grade hardware. By eliminating the dependency on specialized equipment, this design maximizes scalability and broadens community access.

This system will help to improve digital access, social inclusion and independent access to communication for hearing-impaired users.

### **5.3 Sustainability Consideration**

The system being proposed has been designed to run on traditional CPU systems rather than high-end graphics processing units or any other specialized hardware. By eliminating reliance upon external technologies to execute this system, it reduces the amount of energy consumption as well as reducing the dependency on hardware which creates a low-cost, environmentally sustainable solution.

Utilizing open-source tools such as MediaPipe, OpenCV, and Scikit-learn lowers development overhead and provides an adaptable foundation for future software optimization. A major advantage of this system is its reliance solely on a built-in webcam and microphone to track and interpret gestures. By removing any dependence on external electronic hardware or wearable sensors, the design actively reduces potential electronic waste and drastically lowers maintenance complexity for the end-user.

Finally, the architecture from which this system has been designed is scalable, allowing for future growth and integration of new assistive technology solutions, while maintaining the efficient use of resources.

# Chapter 6

## CONCLUSION AND FUTURE SCOPES

### 6.1 Summary of Findings

This developed project, “AI Sign Language to Speech Converter,” successfully demonstrated the development of a lightweight, intelligent, and real-time Indian Sign Language *ISL* recognition system using Artificial Intelligence, Computer Vision, Machine Learning, and Natural Language Processing techniques.

Its primary goal was to eliminate the communication barrier for people in the hearing community who have never been exposed to sign language, by allowing them to use text and even spoken word outputs of gestures made by another person.

The developed system captures real-time hand gestures using a webcam and extracts 21 hand landmarks through the MediaPipe Hands framework. These landmarks can then be used to create feature vectors from which a Random Forest classifier can perform efficient gesture recognition.

The new system also has the potential to use any gesture made by a user to assist multiple languages, rather than having a traditional isolated sign language to output one-to-one messages.

Several intelligent modules were successfully integrated into the system, including:

- Real-time gesture typing
- Stable gesture holding mechanism
- Prediction history filtering
- Spell correction using SymSpell

- Next-word prediction
- Clickable suggestion support
- Interactive smart keyboard interface
- Odia transliteration display
- English to Odia translation
- Real-time bilingual speech synthesis using gTTS

The gesture stabilization mechanism significantly reduced noisy predictions, accidental character generation, and repeated gesture outputs during continuous webcam interaction.

The prediction history filter feature was used to provide better prediction reliability by selecting only reliable predictions from recent images rather than accepting predictions directly from older frames

The integration of SymSpell-based spell correction improved text quality by correcting incorrectly recognized gesture characters caused by temporary hand orientation variations or prediction instability.

By adding a next-word prediction module into the system, it reduces total typing effort required for real-time interaction by providing suggestions for completing words in an appropriate context.

Lastly, the proposed new system provided improved regional accessibility through transliterated Odia text output capabilities and the ability to synthesize Bilingual Odia language speech. Recognized English gesture outputs were converted into transliterated Odia representations such as:

HELLO → Namaskara YES → Hmm

The translated text was then converted into real-time Odia speech using Google Text-to-Speech *gTTS* technology.

The tested Random Forest Classifier provided accurate gesture recognition at an overall accuracy of approximately (94%) with low computational complexity and low latency which makes it suitable for use as an abstract processor; ie, computer system executing this approximation in real time

The MediaPipe landmark based approach provides significantly less processing load compared to image classification deep learning models and does not require the use of expensive GPU hardware or wear- able sensor devices.

Overall, the developed system successfully achieved:

- Real-time Indian Sign Language recognition
- Intelligent AI-assisted gesture typing
- Lightweight CPU-based execution
- Efficient multilingual communication assistance
- Real-time transliteration and speech generation
- Improved communication accessibility for hearing-impaired individuals

The project demonstrates how Artificial Intelligence and Machine Learning can be effectively utilized to create affordable, scalable, and practical assistive communication systems for real-world applications.

## 6.2 Limitations of the Work

Although the proposed system achieved efficient real-time performance and high gesture recognition accuracy, there are many limitations still exist.

- The current system mainly focuses on isolated alphabets gesture recognition rather than continuous sentence-level dynamic gesture recognition.
- Gesture recognition accuracy may decrease under poor lighting conditions, backgrounds , or partial hand occlusions or poor webcam features.
- Certain Indian Sign Language gestures with similar finger orientations may occasionally produce prediction confusions so that repeated predictions occurs.

- The dataset size used for training is relatively limited and may affect generalization capability for different users and environments.
- The current next-word prediction module is based on lightweight predefined vocabulary patterns rather than large-scale language models.
- The system currently supports English text generation and Odia transliterated speech synthesis only.
- OpenCV limitations prevent direct rendering of authentic Unicode Odia script without external font rendering engines.
- Low-resolution webcams or low-end CPU systems may introduce processing latency during continuous real-time interaction.
- The Random Forest classifier, while lightweight and computationally efficient, may not achieve the same large-scale learning capability as advanced deep learning transformer architectures.

Despite these limitations, the developed system successfully demonstrates the feasibility of lightweight AI-powered multilingual gesture communication systems for real-world deployment.

### 6.3 Scope for Future Enhancements

The proposed project can be further extended and enhanced in multiple directions to improve recognition capability, communication intelligence, and practical usability.

Future enhancements may include:

- Development of continuous sentence-level Indian Sign Language recognition systems using temporal sequence learning models.
- Integration of advanced Deep Learning and Transformer-based architectures such as CNN LSTM, GRU, and Vision Transformers for improved gesture recognition accuracy.

- Expansion of multilingual communication support for additional regional Indian languages including Hindi, Bengali, Telugu, Tamil, and Kannada.
- Integration of large language models *LLMs* for advanced contextual sentence generation and intelligent conversation assistance.
- Mobile application deployment for Android and iOS platforms to improve portability and accessibility.
- Web-based cloud deployment for remote real-time communication systems.
- Integration of advanced NLP-based contextual auto-completion and semantic sentence prediction.
- Real-time dynamic sign language sentence translation instead of isolated gesture recognition.
- Enhanced dataset collection involving multiple users, hand orientations, lighting conditions, and regional sign variations for improved generalization capability.
- Support for authentic Unicode Odia rendering using advanced font rendering frameworks.
- GPU acceleration and optimized inference pipelines for ultra-low latency real-time performance.
- Integration with wearable assistive devices and smart healthcare communication systems.
- Voice customization and emotional speech synthesis for improved human-computer interaction.
- AI-based adaptive learning systems capable of learning personalized user gesture styles over time.

In the near future, AI, Computer Vision, NLP, and HCI advancements will allow for significant enhancements in Intelligent Gesture Communication Systems, ultimately providing better, more natural, efficient, and inclusive communication for individuals with hearing impairment.

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# APPENDIX

## Project Proposal

### Introduction

Overview Hearing impaired people's social interactions with those who do not know sign language are negatively affected by the existence of multiple barriers to effective communication (Barney et al., 2009; Earl et al., 2018). Indian Sign Language (ISL) is one of the main forms of communication for hearing-impaired people in India. However, ISL is not widely known. This limits ISL users from receiving an adequate education; experiencing a healthy life; actively engaging in the work force; and communicating effectively on a daily basis.

However, with advances in artificial intelligence (AI), computer vision, machine learning, and natural language processing (NLP), there has been rapid growth of intelligent, real-time gesture recognition systems designed to eliminate the communication barriers between hearing-impaired and hearing individuals.

As a result, the goal of this project is to create an AI based, multilingual, smart gesture keyboard system that recognizes ISL gestures in real-time using an ordinary web cam, and converts them into meaningful text and spoken language output via an AI speech synthesis engine.

In addition to traditional, isolated gesture recognition systems, the proposed system also integrates intelligent communication assistive features such as: real-time gesture typing; stable gesture holding mechanisms; filtering of prediction histories; spell checking; predicting the next word; multilingual transliteration; and producing bilingual spoken output.

The created system will be a lightweight AI driven communication device that can run efficiently on an ordinary CPU-based computer without the need for either dedicated GPUs or glove-type sensors.

## Objectives of the Project

The major objectives of the proposed project are:

- create a way to recognize Indian Sign Language by using AI and Computer Vision
- Capture live hand gestures through a webcam and then find the landmarks of the hands using MediaPipe
- Build a Random Forest ML Classifier to classify the hand gestures.
- Allow for gesture typing without the need for a physical keyboard
- Provide mechanisms for holding stable gestures, so that repeated characters can be reduced
- Implement prediction history filtering techniques to increase gestural recognition robustness
- To integrate SymSpell-based spell correction for improving generated sentence quality.
- Use SymSpell-based spell correction to improve the quality of output sentences
- To generate transliterated Odia text outputs for multilingual communication support.
- Provide transliterated Odia text outputs to support multilingual communication
- To design an interactive smart keyboard interface for efficient user interaction.
- Create a lightweight and affordable assistive communication system for people with hearing impairments.

## Problem Statement

The majority of the sign language recognition systems that exist today rely solely on identifying isolated gestures through a gesture classification process with no advanced communication assistance systems for use in a practical way when interacting in the real world.

Additionally, several of these systems use very costly hardware devices, wearable sensor-based gloves, or high-end GPU-based architectures requiring extensive computational resources running deep learning models.

There are also some existing systems that do not support multiple languages, provide audio for translating regional languages into English or vice versa, have spell-checking features for correcting words being entered, provide next word prediction features while typing, or have reliable gesture stabilization features for providing a smooth flow of gestures in an accurate manner.

The proposed project will address these issues through the development of a lightweight, artificial intelligence-based multilingual smart gesture keyboard that can recognize Indian Sign Language gestures in real-time, provide intelligent typing assistance through transliteration and text-to-speech capabilities in both English and Hindi, and run on basic consumer-grade PC webcam and CPU-based technologies.

## **Proposed Methodology**

The proposed system is made up of many individual interconnected modules that all interact to provide intelligent real-time communication assistance.

The user is captured continuously with live video being transmitted from the camera. The MediaPipe Hands Framework detects and identifies the user hand area and extracts 21 hand landmarks (finger joints, palm location).

The landmarks identified will have their X and Y coordinates normalized, then combined into a feature vector of 42 dimension used by the random forest classifier to classify the user's gestures.

To help establish typing stability and reduce the noise produced by unstable gestures during continuous webcam interaction the system includes mechanisms to hold gesture input stable as well as a history filter to provide the best possible gesture prediction output.

In addition to generating text, the text generated will also go through a spell correction process based on SymSpell as well as a next word prediction module to provide intelligent typing assistance.

The resulting English gesture will then be transformed into an Odia (transliterated) text output and also produce a real-time bilingual text-to-speech output using Google Text-to-Speech (gTTS) service.

The system will also provide interactive UI controls such as Speak, Clear and Stop buttons to allow for efficient user interaction.

## **Tools and Technologies Used**

- Python Programming Language
- OpenCV for webcam handling and visualization
- MediaPipe for hand landmark extraction
- Scikit-learn for Random Forest classification
- SymSpell for spell correction
- gTTS for speech synthesis
- Deep Translator for English-to-Odia translation
- NumPy for numerical computation
- PyCharm IDE for development

## **Expected Outcomes**

The proposed project is expected to achieve:

- Accurate real-time Indian Sign Language recognition
- Lightweight CPU-based execution
- Real-time gesture typing
- Reduced noisy predictions and repeated character generation
- Improved typing stability using gesture holding mechanisms
- Intelligent spell correction and next-word prediction

- Multilingual transliteration support
- Real-time bilingual speech synthesis
- Interactive smart keyboard interface
- Affordable assistive communication support for hearing-impaired individuals

## Applications

The proposed system can be used in multiple real-world applications including:

- Assistive communication systems for hearing-impaired individuals
- Smart educational environments
- Human-computer interaction systems
- Public communication assistance systems
- Healthcare communication support
- AI-based accessibility applications
- Multilingual communication platforms

## Conclusion

The new AI-powered multilingual smart gesture keyboard is intended to fill the void between those with hearing disabilities and people who do not have a hearing impairment by using AI, machine learning, computer vision, and natural language processing technologies.

Real-time gesture recognition, intelligent typing assistance, multilingual transliteration, and bilingual text-to-speech are all combined into one easy-to-use solution that is cost effective, scalable, and feasible as an assistive communication aid in everyday situations. deployment.

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## Team Contribution

| Name                 | Contribution                                                                                                                                                                                                                                           |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bishwa Ranjan Mishra | Developed the gesture recognition system using MediaPipe and Random Forest Classifier                                                                                                                                                                  |
| Ashutosh Sahoo       | Collected and organized the gesture dataset, performed image preprocessing and feature extraction, integrated spell correction and speech synthesis modules, and coordinated overall system implementation and assisted in model training and testing. |
| Saurav kumar Jena    | Implemented multilingual translation and Odia transliteration features, designed the user interface, and contributed to system integration.                                                                                                            |
| Siddhant Mahapatra   | Prepared project documentation, literature survey, experimental analysis, result evaluation, and assisted in debugging and validation of the system.                                                                                                   |

# Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

## Course Outcomes (COs)

- CO1:** Apply engineering knowledge and computing fundamentals to analyze complex real-world problems, identify requirements, and define appropriate project objectives.
- CO2:** Design and develop end-to-end software or computing solutions using appropriate algorithms, architectures, and modern tools to meet specified constraints.
- CO3:** Conduct investigations, experimentation, and performance evaluation using suitable metrics, datasets, simulations, or prototypes to validate project outcomes.
- CO4:** Demonstrate the ability to function effectively as an individual and as a team member/leader, managing tasks, timelines, and resources in multidisciplinary environments.
- CO5:** Communicate technical ideas, design decisions, and results effectively through well-structured project reports, presentations, and demonstrations.
- CO6:** Recognize and apply professional ethics, societal impact, sustainability considerations, and lifelong learning principles in project development and execution.

## Program Outcomes (POs)

- PO1: Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
- PO2: Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
- PO3: Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs

with appropriate consideration for public health and safety, and cultural, societal and environmental considerations.

**PO4: Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.

**PO5: Modern tool usage:** Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

**PO6: The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

**PO7: Environment and sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of, and need for, sustainable development.

**PO8: Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

**PO9: Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.

**PO10: Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large.

**PO11: Project management and finance:** Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work as a member and leader in a team to manage projects in multidisciplinary environments.

## **Program Specific Outcomes (PSOs)**

**PSO1:** The ability to understand and develop computer programs in the areas related to business intelligence, big data analytics, web design and networking for efficient design of computer-based system of varying complexity.

**PSO2:** The ability to apply standard practices and strategy in software project development using open ended programming environments to deliver a quality product for business success.

## Evidences Linked to CO–PO & PSO Mapping

| COs | Mapped POs and PSOs                        | Evidence in Project Work                                                                                                                                                       |                                             |
|-----|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
|     |                                            | Description                                                                                                                                                                    | Section with Evidence                       |
| CO1 | PO1<br>(Engineering Knowledge)             | 1. Applied AI, Machine Learning, and Computer Vision concepts<br>2. Used MediaPipe hand landmark extraction<br>3. Implemented Random Forest classifier for gesture recognition | Chapter 3<br>Section 3.4<br>Section 3.5     |
|     | PO2<br>(Problem Analysis)                  | 1. Identified communication barriers for hearing-impaired users<br>2. Analyzed limitations of existing gesture systems<br>3. Developed lightweight real-time solution          | Chapter 1<br>Section 1.2<br>Chapter 2       |
|     | PSO1                                       | 1. Developed AI-based gesture recognition system<br>2. Implemented intelligent text generation<br>3. Applied machine learning for real-time prediction                         | Chapter 3<br>Section 3.4<br>Section 3.5     |
| CO2 | PO3<br>(Design/Development of Solutions)   | 1. Designed real-time gesture communication system<br>2. Integrated spell correction and speech synthesis<br>3. Developed multilingual smart keyboard interface                | Chapter 4<br>Section 4.3<br>Section 4.4     |
|     | PO5<br>(Engineering Tool Usage)            | 1. Used OpenCV and MediaPipe frameworks<br>2. Used Scikit-learn and SymSpell libraries<br>3. Integrated gTTS and Deep Translator APIs                                          | Chapter 3<br>Section 3.6<br>Chapter 4       |
|     | PSO1                                       | 1. Developed intelligent AI communication platform<br>2. Implemented real-time prediction system<br>3. Integrated multilingual translation modules                             | Chapter 4<br>Section 4.3<br>Section 4.5     |
| CO3 | PO2<br>(Problem Analysis)                  | 1. Evaluated gesture prediction challenges<br>2. Analyzed unstable gesture predictions<br>3. Implemented gesture stabilization mechanisms                                      | Chapter 4<br>Section 4.4<br>Section 4.5     |
|     | PO4<br>(Investigation of Complex Problems) | 1. Conducted model accuracy evaluation<br>2. Used confusion matrix and validation analysis<br>3. Compared prediction stability and latency                                     | Chapter 4<br>Section 4.5<br>Result Analysis |

*Continued on next page*

| COs        | Mapped POs and PSOs                                  | Description                                                                                                                                               | Section with Evidence                                      |
|------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
|            | <b>PO4</b><br>(Investigation of Complex Problems)    | 1. Tested system under real-time conditions<br>2. Evaluated multilingual speech generation<br>3. Measured gesture recognition accuracy                    | Chapter 4<br>Section 4.2<br>Section 4.5                    |
|            | <b>PSO1</b>                                          | 1. Performed real-time AI experimentation<br>2. Conducted model training and testing<br>3. Evaluated intelligent prediction modules                       | Chapter 4<br>Section 4.3<br>Section 4.5                    |
|            | <b>PSO2</b>                                          | 1. Developed complete software pipeline<br>2. Integrated real-time communication modules<br>3. Validated final AI-based application                       | Chapter 4<br>Section 4.3<br>Section 4.5                    |
| <b>CO4</b> | <b>PO8</b><br>(Individual & Collaborative Team Work) | 1. Performed collaborative project development<br>2. Shared implementation and documentation tasks<br>3. Coordinated model training and testing           | Team Contribution<br>Appendix<br>Project Development       |
|            | <b>PO10</b><br>(Project Management & Finance)        | 1. Managed project timeline and implementation phases<br>2. Developed low-cost CPU-based solution<br>3. Organized dataset collection and testing          | Chapter 1<br>Chapter 4<br>Appendix                         |
|            | <b>PSO2</b>                                          | 1. Implemented modular software architecture<br>2. Developed scalable AI application<br>3. Managed end-to-end project workflow                            | Code Structure<br>Chapter 4<br>Appendix                    |
| <b>CO5</b> | <b>PO9</b><br>(Communication)                        | 1. Prepared technical documentation and report<br>2. Presented system architecture and workflow<br>3. Demonstrated project outputs and analysis           | Entire Report<br>System Architecture<br>Results<br>Section |
| <b>CO6</b> | <b>PO6</b><br>(The Engineer and the World)           | 1. Developed assistive communication technology<br>2. Improved accessibility for hearing-impaired users<br>3. Addressed real-world communication barriers | Chapter 1<br>Chapter 5<br>Conclusion                       |
|            | <b>PO7</b><br>(Ethics)                               | 1. Used ethical AI-based communication support<br>2. Promoted inclusive accessibility solutions<br>3. Ensured affordable and lightweight deployment       | Chapter 5<br>Ethical Considerations<br>Conclusion          |

*Continued on next page*

| COs | Mapped POs and PSOs                 | Description                                                                                                                            | Section with Evidence                  |
|-----|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
|     | <b>PO11</b><br>(Life-Long Learning) | 1. Learned modern AI and NLP technologies<br>2. Explored multilingual speech systems<br>3. Applied advanced Computer Vision techniques | Chapter 3<br>Chapter 4<br>Future Scope |

## Code Structure

The proposed AI Sign Language to Speech Converter system was developed using a modular programming approach in Python. Each module is responsible for a specific functionality such as dataset generation, gesture recognition, machine learning model training, spell correction, multilingual translation, and speech synthesis.

The complete codebase is organized into multiple Python files for better maintainability, readability, and modularity.

| File Name                          | Purpose / Description                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| dataset_creation.py                | Captures hand gesture images using webcam and creates the custom gesture dataset for training.                             |
| create_dataset.py                  | Processes captured gesture images and extracts MediaPipe hand landmarks for feature generation.                            |
| model_creation.py                  | Trains the Random Forest classifier using extracted landmark feature vectors and gesture labels.                           |
| project.py<br>hand_detection.py    | Main execution file responsible for real-time gesture recognition, text generation, transliteration, and speech synthesis. |
| model.p                            | Serialized trained Random Forest model stored using Pickle for real-time prediction.                                       |
| frequency_dictionary_en_82_765.txt | Dictionary file used by the SymSpell spell correction module.                                                              |
| voice.mp3                          | Temporary speech output file generated during Odia speech synthesis using gTTS.                                            |
| requirements.txt                   | Contains all Python libraries and dependencies required to execute the project.                                            |

Table 6.2: Code Structure of the Proposed System

By implementing a modular system, it creates an environment for easier scalability and will simplify debugging, future changes, and feature integration. Each module is designed to handle one specific task alone; collectively, they will work together to produce an intelligent real-time gesture communication network.

# Code Listings

The major code implementations used in the proposed AI-powered multilingual gesture keyboard system are summarized below.

## 1. Dataset Creation Module

This module captures real-time hand gesture images using a webcam and stores them in separate folders corresponding to gesture classes.

```
cap = cv2.VideoCapture(0)

while True:

    ret, frame = cap.read()

    cv2.imshow("frame", frame)
```

The dataset contains gesture images for alphabets  $A - Z$  and digits  $0 - 9$ .

## 2. MediaPipe Landmark Extraction

MediaPipe Hands framework is used to extract 21 hand landmarks from detected hands.

```
mp_hands = mp.solutions.hands

hands = mp_hands.Hands(
    static_image_mode=False,
    max_num_hands=2
)
```

The extracted landmark coordinates are normalized and converted into feature vectors.

## 3. Random Forest Model Training

The extracted hand landmark features are used to train the Random Forest classifier.

```
model = RandomForestClassifier(
```

```

        n_estimators=100,
        random_state=42
    )

model.fit(x_train, y_train)

```

The trained model is saved using Pickle for real-time prediction.

## 4. Real-Time Gesture Prediction

The trained classifier predicts gestures from live webcam frames.

```

prediction = model.predict(
    [np.asarray(data_aux)]
)[0]

```

Predicted gestures are converted into text outputs.

## 5. Stable Gesture Holding Mechanism

To prevent repeated character generation, the system appends characters only after stable gesture detection.

```

if elapsed > 6:

    text += stable_prediction

```

This mechanism improves typing reliability and reduces noisy predictions.

## 6. Spell Correction using SymSpell

The SymSpell algorithm corrects incorrectly predicted gesture characters.

```

suggestions = sym_spell.lookup(
    last_word,
    Verbosity.TOP,
    max_edit_distance=2
)

```

This improves sentence quality and communication accuracy.

## 7. Next Word Prediction

Frequently used phrase patterns are utilized to generate next-word suggestions.

```
common_phrases = {  
  
    "hello": ["world", "there"]  
  
}
```

The module improves communication efficiency.

## 8. Odia Speech Synthesis

Recognized English text is translated into Odia and converted into speech using gTTS.

```
tts = gTTS(  
    text=translated,  
    lang='hi'  
)  
  
tts.save(filename)
```

The generated speech is played in real time.

## 9. Interactive Smart Keyboard Interface

The system integrates interactive buttons such as Speak, Clear, and Stop.

```
cv2.rectangle(  
    frame,  
    btn_speak[:2],  
    btn_speak[2:],  
    (0,255,0),  
    -1  
)
```

The interface provide efficient user interaction and instant communication assistance.

The complete implementation is the combination of Computer Vision, Machine Learning, Natural Language Processing and multilingual speech technologies which will help to make lightweight AI-powered assistive communication platform.